Benchmark: **MA.7.GR.1.4** · Area of a circle Student code: _____ Calculator: yes

Learning goal: I can find the area of a circle. (circle your level after scoring)

4	I can solve a real-world problem using circumference and area together, including semicircles.
3	I can find the area of a circle from its radius or diameter.
2	I can find the radius from the diameter and square a number.
1	I need help to start this benchmark.

1. A circle has a diameter of 12 meters.

What is the area of the circle? Write your answer in terms of π.

2. A circular garden has a diameter of 14 feet.

What is the area of the circular garden? Round your answer to the nearest hundredth. Use 3.14 for π.

3. A 24-inch pizza is cut into 9 equal slices. (The size of a pizza is its diameter.) A student eats 5 slices.

How many square inches of pizza did the student eat? Write your answer in terms of π.

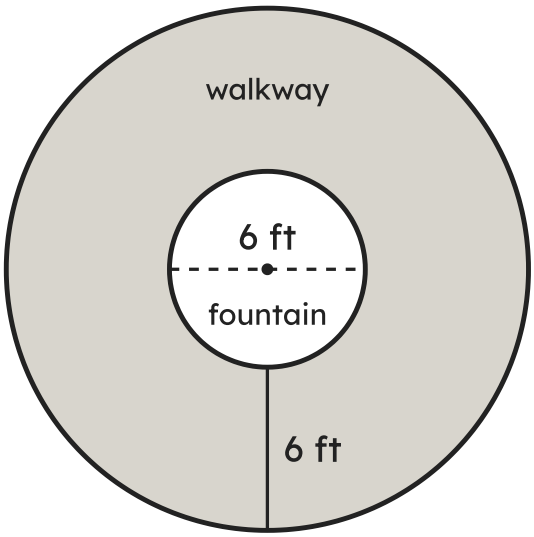
Turn over →

4. A circular skating rink has a radius of 21 meters.

What is the area of the circular skating rink? Round your answer to the nearest hundredth. Use 3.14 for π.

5. A circular fountain has a diameter of 6 feet. A walkway 6 feet wide surrounds the fountain, as shown.

What is the area of the walkway, in square feet?



Not drawn to scale

A. 72π

B. 36π

C. 108π

D. 81π

Score: ____ / 5 **My level: 1 2 3 4** **→ copy both to your progress strip**

Answer key — C-GR.1.4 · MA.7.GR.1.4 · Area of a circle

Unit 4: Area. Regular = on-benchmark item; Rigor = the item that needs the benchmark's reasoning. Mastery on the weekly check = 4 of 5.

Weekly check (5 items)

1. Regular **36π square meters**

The radius is $12 \div 2 = 6$ meters.

$A = \pi r^2 = \pi \times 6^2 = 36\pi$ square meters.

2. Regular **153.86 square feet**

The radius is $14 \div 2 = 7$ feet.

$A = \pi r^2 = \pi \times 7^2 = 49\pi$.

With 3.14 for π: $49 \times 3.14 = 153.86$ square feet.

3. Regular **80π square inches**

The diameter is 24 inches, so the radius is $24 \div 2 = 12$ inches and the whole pizza has area $\pi(12)^2 = 144\pi$ square inches.

The student ate $\frac{5}{9}$ of the pizza.

$\frac{5}{9} \times 144\pi = 80\pi$ square inches.

4. Regular **1384.74 square meters**

$A = \pi r^2 = \pi \times 21^2 = 441\pi$.

With 3.14 for π: $441 \times 3.14 = 1384.74$ square meters.

5. Rigor **A.** 72π

The fountain's radius is $6 \div 2 = 3$ feet, so the outer edge of the walkway is $3 + 6 = 9$ feet from the center.

Fountain and walkway together: $\pi(9)^2 = 81\pi$.

Fountain alone: $\pi(3)^2 = 9\pi$.

Walkway: $81\pi - 9\pi = 72\pi$ square feet.

The walkway is a ring around the fountain, not a circle with a radius of 6 feet.

Re-check strip (3 items, C-GR.1.4-R)

1. Regular **196π square inches**

The radius is $28 \div 2 = 14$ inches.

$A = \pi r^2 = \pi \times 14^2 = 196\pi$ square inches.

2. Regular **803.84 square inches**

$A = \pi r^2 = \pi \times 16^2 = 256\pi$.

With 3.14 for π: $256 \times 3.14 = 803.84$ square inches.

3. Regular **75π square inches**

The diameter is 30 inches, so the radius is $30 \div 2 = 15$ inches and the whole pizza has area $\pi(15)^2 = 225\pi$ square inches.

The student ate $\frac{1}{3}$ of the pizza.

$\frac{1}{3} \times 225\pi = 75\pi$ square inches.

Comparison strip (2 items, C-GR.1.4-X) — used on a later week's re-check when this benchmark was *not* retaught

1. Regular **B.** 484π

$A = \pi r^2 = \pi \times 22 \times 22 = 484\pi$.

The value 44π would be the circumference instead.

2. Regular **2289.06 square meters**

$A = \pi r^2 = \pi \times 27^2 = 729\pi$.

With 3.14 for π: $729 \times 3.14 = 2289.06$ square meters.

Reteach plan — C-GR.1.4

MA.7.GR.1.4 · Area of a circle

Item that gave the most trouble (from both tally sheets):

What the mistake looks like:

Worked example for Tuesday (8–10 min):

Three practice items (one change at a time):

Decision line: Reteach on _____ Huddle: in person / paper Date: _____ Re-check given on _____

7th Grade Math · Re-check (3 items)

C-GR.1.4-R

Benchmark: **MA.7.GR.1.4** · Area of a circle

Student code: _____ Calculator: yes

1. A circle has a diameter of 28 inches.

What is the area of the circle? Write your answer in terms of π.

2. A round tabletop has a radius of 16 inches.

What is the area of the round tabletop? Round your answer to the nearest hundredth. Use 3.14 for π.

3. A 30-inch pizza is cut into 3 equal slices. (The size of a pizza is its diameter.) A student eats 1 slice.

How many square inches of pizza did the student eat? Write your answer in terms of π.

Score: ____ / 3 → **copy to your progress strip**

7th Grade Math · Re-check (3 items)

C-GR.1.4-R

Benchmark: **MA.7.GR.1.4** · Area of a circle

Student code: _____ Calculator: yes

1. A circle has a diameter of 28 inches.

What is the area of the circle? Write your answer in terms of π.

2. A round tabletop has a radius of 16 inches.

What is the area of the round tabletop? Round your answer to the nearest hundredth. Use 3.14 for π.

3. A 30-inch pizza is cut into 3 equal slices. (The size of a pizza is its diameter.) A student eats 1 slice.

How many square inches of pizza did the student eat? Write your answer in terms of π.

Score: ____ / 3 → **copy to your progress strip**

7th Grade Math · Comparison items (2) C-GR.1.4-X

Benchmark: **MA.7.GR.1.4** · Area of a circle Student code: _____ Calculator: yes

1. A circle has a radius of 22 feet.

What is the area of the circle, in square feet?

A. 44π

B. 484π

C. 1936π

D. 22π

2. A circular skating rink has a radius of 27 meters.

What is the area of the circular skating rink? Round your answer to the nearest hundredth. Use 3.14 for π.

Score: ____ / 2 → **copy to your progress strip**

7th Grade Math · Comparison items (2) C-GR.1.4-X

Benchmark: **MA.7.GR.1.4** · Area of a circle Student code: _____ Calculator: yes

1. A circle has a radius of 22 feet.

What is the area of the circle, in square feet?

A. 44π

B. 484π

C. 1936π

D. 22π

2. A circular skating rink has a radius of 27 meters.

What is the area of the circular skating rink? Round your answer to the nearest hundredth. Use 3.14 for π.

Score: ____ / 2 → **copy to your progress strip**